ANSWERS – EE10A 2017

1. Use of basic principles to understand the use of a pn diode as a temperature sensor. New application not seen before by the students but based on fundamental knowledge of semiconductor devices applied to pn junctions.

a) [5]

From formulae sheet:

$$p = N_V \, exp\left(\frac{E_V - E_F}{kT}\right)$$

$$n = N_C \, exp\left(\frac{E_F - E_C}{kT}\right)$$

definition of intrinsic energy E_i:

$$p_i = n_i = N_V \, exp\left(\frac{E_V - E_i}{kT}\right)$$

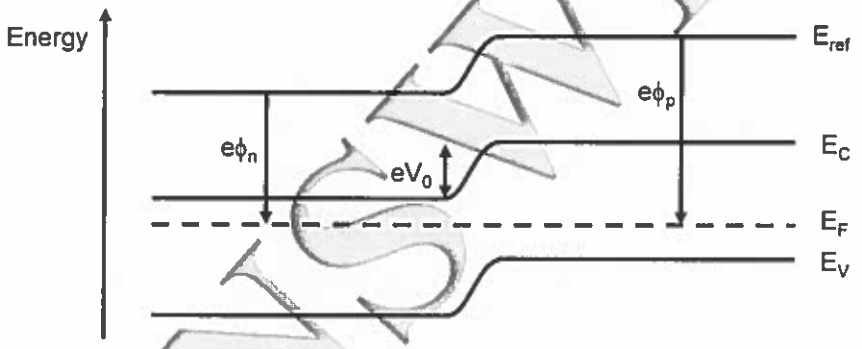
$$n_i = N_C \, exp\left(\frac{E_i - E_C}{kT}\right)$$

product of these two equations:

$$n_i^2 = N_C \, N_V exp\left(\frac{E_i - E_C}{kT}\right) exp\left(\frac{E_V - E_i}{kT}\right) = N_C \, N_V \, exp\left(\frac{E_V - E_C}{kT}\right) = N_C N_V exp\left(\frac{-E_G}{kT}\right)$$

b). [5]

Draw energy band diagram:



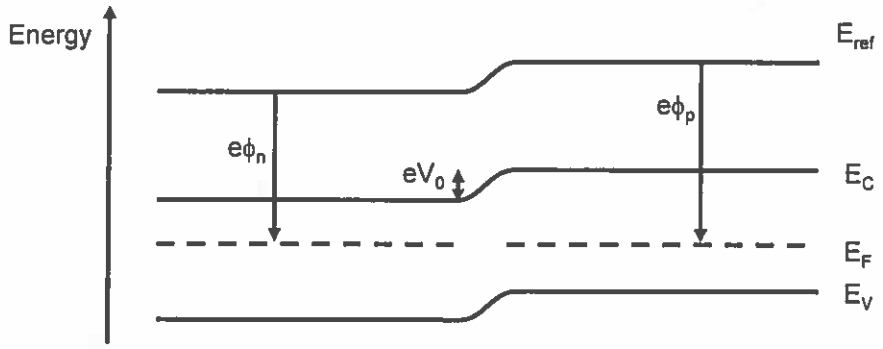
ϕ_n and ϕ_p are the workfunction of n and p type region, respectively.

From the plot it is easy to see that: $V_0 = \phi_p - \phi_n$

c). [7]

From the solution in a) we can see that the intrinsic carrier concentration is increasing with increasing temperature, assuming that the bandgap and the effective density of state vary more slowly than the exponential function of T. The density of ionized doping atoms will not vary much for temperatures around room temperature (nearly complete ionization at room temperature). This means that with increasing T and thus increasing n_i, the effective carrier concentration will be more and more determined by n_i rather than the doping. Thus the Fermi levels will move closer to the intrinsic level. This makes the energy band diagram in b) change to the plot below for higher temperatures. [4]

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Thus the workfunctions change thus the build-in voltage decreases.
When the build in voltage decreases then less voltage needs to be applied across the diode to lower the potential barrier eV_0 to switch the diode on. Thus for T increase V_{on} decreases.
[3]

d) bookwork with a twist (explicit influence of temperature is new) and without giving the switching circuit as done in previous exam questions. This thus also tests the students' knowledge on terminology. [8]

Assume that the delay is mainly caused by the minority carrier holes (derivation is same when assuming this for electrons).

From coursenotes derive switching equations:

$$\frac{dQ_p(t)}{dt} = i_p(t) - \frac{Q_p(t)}{\tau_p}$$

For $0 < t < t_{sd}$ with t_{sd} the storage delay time, we know that i_p remains constant at $-I_r$, thus:

$$\frac{dQ_p(t)}{dt} = -I_r - \frac{Q_p(t)}{\tau_p}$$

Separation of variables:

$$\frac{dQ_p(t)}{Q_p(t) + \tau_p I_r} = -\frac{dt}{\tau_p}$$

Integrate:

$$\int_{Q_p(0)}^{Q_p(t)} \frac{dQ_p(t)}{Q_p(t) + \tau_p I_r} = -\int_0^t \frac{dt}{\tau_p}$$

$$\ln\left(Q_p(t) + \tau_p I_r\right)\Big|_{\tau_p I_F}^{Q_p(t)} = -\frac{t}{\tau_p}$$

$$\ln\left(Q_p(t) + \tau_p I_r\right) - \ln\left(\tau_p I_F + \tau_p I_r\right) = -\frac{t}{\tau_p}$$

$$\ln\frac{\left(Q_p(t) + \tau_p I_r\right)}{\left(\tau_p I_F + \tau_p I_r\right)} = -\frac{t}{\tau_p}$$

$$Q_p(t) = -\tau_p I_r + \left(\tau_p I_F + \tau_p I_r\right)\exp\left(-\frac{t}{\tau_p}\right)$$

or

$$Q_p(t) = \tau_p\left[-I_r + \left(I_f + I_r\right)\exp\left(\frac{-t}{\tau_p}\right)\right]$$

valid for $0 < t < t_{sd}$ only.

SOLUTIONS

Approximate that at $t=t_{sd}$, $Q_p(t_{sd})=0$. Thus:

$$0 = \tau_p \left[-I_r + (I_f + I_r) \exp\left(\frac{-t_{sd}}{\tau_p} \right) \right]$$

The storage delay time is then:

$$t_{sd} \cong \tau_p \ln\left[1 + \frac{I_f}{I_r} \right]$$ [4] marks for the derivation

Calculating t_{sd} at T = 25 °C and 75 °C using I_f from figure 1.1 at diode voltage of 0.45V.
[Note: the IVs are plotted such that these values can be easily extracted with a ruler.]

T = 25 °C I_f = 100 µA - $t_{sd}(\mu s) = 1 \times ln\left[1 + \frac{100}{200}\right] = 0.41\mu s$

T = 75 °C I_f = 1000 µA - $t_{sd}(\mu s) = 1 \times ln\left[1 + \frac{1000}{200}\right] = 1.79\mu s$

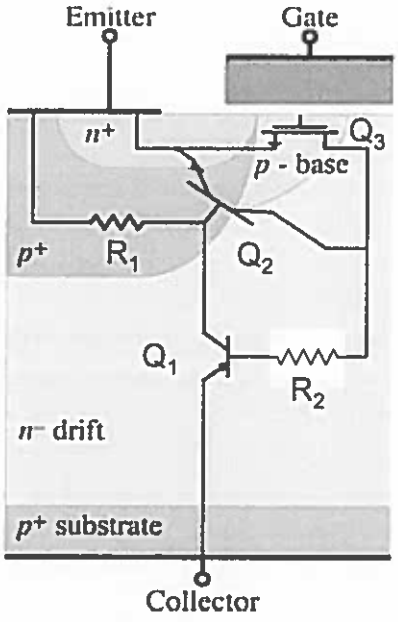
The storage delay time increases by almost a factor of 3 (1.93) [4]

[Note: this will cause even more heating of the diode (thermal runaway).]

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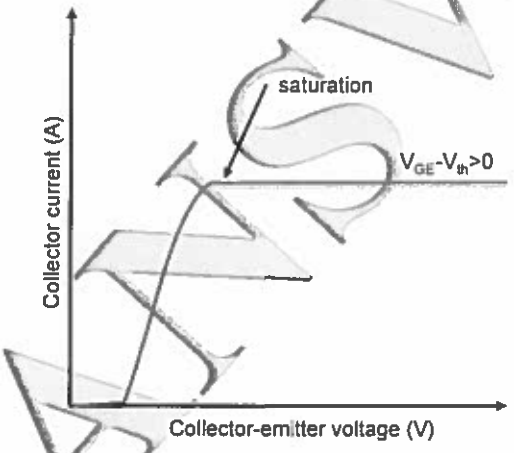
2. This question is bookwork

a) A half cell is drawn below. [8]



b) main IGBT components are Q3: MOSFET for control of base current (input) and Q1 pnp BJT for output current generation. [8]
The parasitics are: Q2 across the MOSFET, this is normally off by shorting the p+ (can also be made p) and n+ region via the top contact, however a voltage drop across the bulk resistor R1 can create a voltage difference turning the parasitic BJT on. R2 is the parasitic resistor from the lowly doped n- drift region.

c) Saturation is due to pinch-off of the MOSFET Q3. [4]



d) A typical transient characteristic of an IGBT is given in fig. 2.1. Answer the following questions.

i) charging of the gate capacitance until the threshold voltage is reached. [2]

ii) removal of the excess minority carrier charge in the drift region via recombination. [2]

iii) In the EE2-10A module loads are resistive and therefore no overshoots are observed. Overshoots occur when loads are inductive. [1]

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